If statement

In [14]:
```
cars = ['audi', 'bmw', 'toyota', 'subaru']
for car in cars:
    if car=='bmw':
        print(car.upper())
    else:
        print(car.title())
```

```
Audi
BMW
Toyota
Subaru
```

Logical test--True / False

In [17]:
```
car=='subaru'
```

Out[17]:
```
True
```

In [18]:
```
car=='iphone'
```

Out[18]:
```
False
```

Checking for Inequality

In [19]:
```
requested_topping='mushroom'
if requested_topping != 'anchovies':
    print("Hold the anchovies.")
```

```
Hold the anchovies.
```

Numerical Comparisons

In [20]:
```
age=18
age==18
```

Out[20]:
```
True
```

In [21]:
```
age=18
age>21
```

Out[21]:
```
False
```

In [25]:
```
age<19 and age >21
```

Out[25]:
```
False
```

In [26]:
```
answer=17
if answer !=42:
```

```
    print("That is not the correct answer. Please try again!")
```

```
That is not the correct answer. Please try again!
```

and / or--Checking Multiple Conditions

In [27]:
```
# and
age_0=22
age_1=18
age_0>=21 and age_1 >=21
```

Out[27]:
```
False
```

In [29]:
```
age_0=22
age_1=23
age_0>=21 and age_1 >=21
```

Out[29]:
```
True
```

In [14]:
```
# or
age_0=22
age_1=23
age_0>=21 or age_1 >=21
```

Out[14]:
```
True
```

Checking in a List

In [30]:
```
requested_toppings = ['mushrooms', 'onions', 'pineapple']
'mushrooms' in requested_toppings
```

Out[30]:
```
True
```

In [32]:
```
requested_toppings = ['mushrooms', 'onions', 'pineapple']
'tomato' in requested_toppings
```

Out[32]:
```
False
```

Checking not in a List

In [33]:
```
requested_toppings = ['mushrooms', 'onions', 'pineapple']
'mushrooms' not in requested_toppings
```

Out[33]:
```
False
```

In [35]:
```
banned_users = ['andrew', 'carolina', 'david']
user='marie'

if user not in banned_users:
    print(user.title()+", you can post a response if you wish.")
```

```
Marie, you can post a response if you wish.
```

Boolean Expressions

In [36]:
```
game_active=True
can_edit=False
```

In [22]:
```
# 5-1
food='beef'
print("Is food=='beef'? I predict True.")
print(food=='beef')
```

```
Is food=='beef'? I predict True.
True
```

In [23]:
```
food='vegetable'
print("Is food=='beef'? I predict False.")
print(food=='beef')
```

```
Is food=='beef'? I predict True.
False
```

In [24]:
```
food='vegetable'
print("Is food!='beef'? I predict True.")
print(food!='beef')
```

```
Is food!='beef'? I predict True.
True
```

In [25]:
```
car='bmw'
car=='bmw'
```

Out[25]:
```
True
```

In [26]:
```
car='BMW'
car.lower()=='bmw'
```

Out[26]:
```
True
```

In [33]:
```
21==21 or 21<21
```

Out[33]:
```
True
```

In [34]:
```
21>=21 or 18>12
```

Out[34]:
```
True
```

In [35]:
```
21>=21 and 18>120
```

Out[35]:
```
False
```

if

In [37]:
```
cars=['bmw','Audi','YAMAHA']
car='BMW'
if car.lower() in cars:
    print("We have the car "+car.title())
```

```
else:
    print("We don't have the car!")
```

```
We have the car Bmw
```

In [38]:
```
cars=['bmw','Audi','YAMAHA']
car='xiaomi'
if car.lower() not in cars:
    print("We don't have the car!")
```

```
We don't have the car!
```

In [47]:
```
age=19
if age >=18:
    print("You are old enough to vote!")
```

```
You are old enough to vote!
```

if-else

In [49]:
```
age=11
if age >=18:
    print("You are old enough to vote!")
else:
    print("Sorry,you can not vote.")
```

```
Sorry,you can not vote.
```

if-elif

In [6]:
```
age=89
if age<4:
    price=0
elif age<18:
    price=5
elif age<60:
    price=10
elif age<100:
    price=15

print("Your admission cost is $"+str(price)+".")
```

```
Your admission cost is $15.
```

if-elif-else

In [39]:
```
age=12
if age<4:
    print("Your admission cost is $0.")
elif age <18:
    print("Your admission cost is $5.")
else:
    print("Your admission cost is $10.")
```

```
Your admission cost is $5.
```

In [55]:
```
age=19
if age<4:
    price=0
elif age<18:
    price=5
else:
    price=10

print("Your admission cost is $"+str(price)+".")
```

```
Your admission cost is $10.
```

In [4]:
```
age=69
if age<4:
    price=0
elif age<18:
    price=5
elif age<60:
    price=10
else:
    price=15

print("Your admission cost is $"+str(price)+".")
```

```
Your admission cost is $15.
```

单 if--Testing Multiple Conditions

In [11]:
```
requested_toppings=['mushroom','extra cheese']

if 'mushroom' in requested_toppings:
    print("Adding mushroom.")

if 'pepperoni' in  requested_toppings:
    print("Adding pepperoni.")

if 'extra cheese' in  requested_toppings:
    print("Adding extra cheese")

print("\nFinished making your pizza.")
```

```
Adding mushroom.
Adding extra cheese

Finished making your pizza.
```

Excercise

In [41]:
```
# 5-3 if
alien_color='green'
if alien_color=='green':
    print("The player just earned 5 points")
```

```
The player just earned 5 points
```

In [19]:
```
# 5-3 if
alien_color='yellow'
```

```
if alien_color=='green':
    print("The player just earned 5 points")
```

In [24]:
```
# 5-4 if-else
alien_color='red'
if alien_color=='green':
    print("The player just earned 5 points")
else:
    print("The player just earned 10 points")
```

```
The player just earned 10 points
```

In [42]:
```
# 5-5 if-elif-else
alien_color='red'
if alien_color=='green':
    print("The player just earned 5 points.")
elif alien_color=='yellow':
    print("The player just earned 10 points.")
else:
    print("The player just earned 15 points.")
```

```
The player just earned 15 points.
```

In [43]:
```
# 5-6 解法1 if-elif-else
age=64
if age<2:
    stage='baby'
elif 2<= age <4:
    stage='toddler'
elif 4<= age <13:
    stage='kid'
elif 13<= age <20:
    stage='teenager'
elif 20<= age <65:
    stage='adult'
else:
    stage='elder'

print("The person is an "+stage+".")
```

```
The person is an adult.
```

In [44]:
```
# 5-6 解法2 if-elif-else
age=64
if age<2:
    print("The person is a baby.")
elif 2<= age <4:
    print("The person is a toddler.")
elif 4<= age <13:
    print("The person is a kid.")
elif 13<= age <20:
    print("The person is a teenager.")
elif 20<= age <65:
    print("The person is an adult.")
else:
     print("The person is an elder.")
```

```
The person is an adult.
```

In [45]:
```
# 5-7 单if 多重判断
favorite_fruits=['apple','banana','watermelon']
```

```
if 'apple' in favorite_fruits:
    print("You really like apple!")
if 'mango' in favorite_fruits:
    print("You really like mango!")
if 'banana' in favorite_fruits:
    print("You really like banana!")
if 'orange' in favorite_fruits:
    print("You really like orange!")
if 'watermelon' in favorite_fruits:
    print("You really like watermelon!")
```

```
You really like apple!
You really like banana!
You really like watermelon!
```

if + list

In [48]:
```
requested_toppings = ['mushrooms', 'green peppers', 'extra cheese']
for requested_topping in requested_toppings:
    print("Adding " + requested_topping + ".")

print("\nFinished making your pizza!")
```

```
Adding mushrooms.
Adding green peppers.
Adding extra cheese.

Finished making your pizza!
```

In [49]:
```
requested_toppings = ['mushrooms', 'green peppers', 'extra cheese']
for requested_topping in requested_toppings:
    if requested_topping=='green peppers':
        print("Sorry,we are out of green peppers right now.")
    else:
        print("Adding " + requested_topping + ".")

print("\nFinished making your pizza!")
```

```
Adding mushrooms.
Sorry,we are out of green peppers right now.
Adding extra cheese.

Finished making your pizza!
```

if + epmty list

In [49]:
```
usernames=[]  # 空list，不能写成usernames=[‘’]
if usernames:   # 如果usernames list 非空时执行 if语句，是空的话，执行else语句
    for username in usernames:
        if username=='admin':
            print("Hello admin, would you like to see a status report?")
        else:
            print("Hello " + username.title() +", thank you for log-ging again."
else:
    print("We need to find some users!")
```

```
We need to find some users!
```

In [84]:
```
requested_toppings=[]

if requested_toppings:
    for requested_topping in requested_toppings:
        print("Adding " + requested_topping + ".")
    print("\nFinished making your pizza!")

else:
    print("Are you sure you want a plain pizza?")
```

```
Are you sure you want a plain pizza?
```

if + Multiple Lists

In [50]:
```
available_toppings = ['mushrooms', 'olives', 'green peppers','pepperoni', 'pinea
requested_toppings = ['mushrooms', 'french fries', 'extra cheese']

for requested_topping in requested_toppings:
    if requested_topping in available_toppings:
        print("Adding " + requested_topping +".")
    else:
        print("Sorry, we don't have " + requested_topping + ".")

print("\nFinished making your pizza!")
```

```
Adding mushrooms.
Sorry, we don't have french fries.
Adding extra cheese.

Finished making your pizza!
```

Excercise

In [51]:
```
# 5-8:
usernames=['admin','mica','donna','maria','anna','lisa']

for username in usernames:
    if username=='admin':
        print("Hello admin, would you like to see a status report?")
    else:
        print("Hello " + username.title() +", thank you for log-ging again.")
```

```
Hello admin, would you like to see a status report?
Hello Mica, thank you for log-ging again.
Hello Donna, thank you for log-ging again.
Hello Maria, thank you for log-ging again.
Hello Anna, thank you for log-ging again.
Hello Lisa, thank you for log-ging again.
```

5-9 if + empty list

In [52]:
```
# 5-9:薄弱题
usernames=[]
if usernames:
```

```
    for username in usernames:
        if username=='admin':
            print("Hello admin, would you like to see a status report?")
        else:
            print("Hello " + username.title() +", thank you for log-ging again."
else:
    print("We need to find some users!")
```

```
We need to find some users!
```

5-10 if + 多list + lower/upper()

In [54]:

```
# 5-10 薄弱题
current_users=['admin','mica','DONNA','maria','ANNA']
new_users=['lily','anna','donna','jack','MICA']

# 关键转换
current_users_lower=[user.lower() for user in current_users]

for new_user in new_users:
    if new_user.lower() in current_users_lower:
        print(new_user+" has already been used,you need to enter a new username!
    else:
        print(new_user+" the username is available.")
```

```
lily the username is available.
anna has already been used,you need to enter a new username!
donna has already been used,you need to enter a new username!
jack the username is available.
MICA has already been used,you need to enter a new username!
```

5-11 if-elif-else + range()

In [56]:

```
# 5-11
nums=list(range(1,11))
for num in nums:
    if num==1:
        print("1st")
    elif num==2:
        print("2nd")
    elif num==3:
        print("3rd")
    else:
        print(str(num)+"th")
```

```
1st
2nd
3rd
4th
5th
6th
7th
8th
9th
10th
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]: